

Mathematical Tools.

1 Factorial Notation (!)

$$6! = 6 \times 5 \times 4 \times 3 \times 2 \times 1 = 720$$

$$5! = 5 \times 4 \times 3 \times 2 \times 1 = 120$$

$$4! = 4 \times 3 \times 2 \times 1 = 24$$

$$3! = 3 \times 2 \times 1 = 6$$

$$2! = 2 \times 1 = 2$$

$$1! = 1$$

$$6! = 6 \times 5! = 6 \times (6-1)!$$

$$\therefore n! = n \times (n-1)!$$

Vimp Prove that $0! = 1$

we know that

$$n(n-1)! = n!$$

put $n = 1$

$$1(1-1)! = 1!$$

$$1(0)! = 1!$$

$$0! = 1$$

$$(i) \quad \frac{6!}{5!} = \frac{6 \times 5!}{5!} = 6$$

$$(ii) \quad \frac{7!}{5!} = \frac{7 \times 6 \times 5!}{5!} = 42$$

$$(iii) \quad \frac{100!}{98!} = \frac{100 \times 99 \times 98!}{98!} = 9900$$

$$(iv) \quad 3! + 2! = 6 + 2 = 8$$

$$(v) \quad 5! - 3! = 120 - 6 = 114$$

vi

$$4! \times 2! = 24 \times 2 = 48.$$

Q

Evaluate

$$\frac{7!}{6!} + \frac{5!}{3!} + \frac{3!}{5!}$$

$$\frac{7 \times 6!}{6!} + \frac{5 \times 4 \times 3!}{3!} + \frac{3!}{5 \times 4 \times 3!}$$

$$= 7 + 20 + \frac{1}{20}$$

$$= 27 + \frac{1}{20} = \frac{540+1}{20} = \frac{541}{20}$$

Combination.

$${}^nC_r \quad [n \geq r]$$

$${}^nC_r = \frac{n!}{r!(n-r)!}$$

$$\text{e.g.; } \rightarrow \frac{3!}{2! \times (3-2)!} = \frac{6}{2 \times 1} = \frac{6}{2} = 3.$$

$$\rightarrow {}^5C_3 = \frac{5!}{3! \times (5-3)!} = \frac{120}{6 \times 2} = \frac{120}{12} = 10.$$

$$\rightarrow {}^4C_2 = \frac{4!}{2! \times (4-2)!} = \frac{24}{2 \times 2!} = \frac{24}{4} = 6.$$



$$\Rightarrow {}^nC_0 = \frac{n!}{0!(n-0)!} = \frac{n!}{1(n)!} = \frac{n!}{n!} = 1$$

$$\Rightarrow {}^nC_1 = \frac{n!}{1(n-1)!} = \frac{n!}{(n-1)!} = \frac{n(n-1)!}{(n-1)!} = n$$

$$\Rightarrow {}^nC_n = \frac{n!}{n!(n-n)!} = \frac{n!}{n!(0!)} = \frac{n!}{n! \times 1} = 1$$

Q. Evaluate $= {}^2C_0 + {}^5C_1 + {}^3C_3$

$$1 + 5 + 1 = 7$$

out of three students two students are to be selected for a quiz program. what is the no. of possible combinations?

$$n=3, r=2$$

$${}^nC_r = {}^3C_2 = \frac{3!}{2!(3-2)!} = \frac{3!}{2!(1)!}$$

$$= \frac{6}{2 \times 1} = \frac{6}{2} = 3$$

out of 5 students 2 are to be selected for playing a game. what's the no. of possible combinations.

$$n=5, r=2$$

$${}^nC_r = {}^5C_2 = \frac{5!}{2!(5-2)!} = \frac{5!}{2! \times 3!}$$

$$= \frac{120}{2 \times 6} = \frac{120}{12} = 10$$

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Binomial Theorem.
Binomial Theorem for any index.

$$(1+z)^n = 1 + nz + \frac{n(n-1)}{2!} z^2 + \frac{n(n-1)(n-2)}{3!} z^3 + \frac{n(n-1)(n-2)(n-3)}{4!} z^4 + \frac{n(n-1)(n-2)(n-3)(n-4)}{5!} z^5 + \dots$$

Prove B.T.

$$(1+z)^n = 1 + nz + \frac{n(n-1)}{2!} z^2 + \frac{n(n-1)(n-2)}{3!} z^3 + \dots$$

Put $n = 2$.

$$\begin{aligned} (1+z)^2 &= 1 + 2z + \frac{2(2-1)}{2!} z^2 + \frac{2(2-1)(2-2)}{3!} z^3 + \dots \\ &= 1 + 2z + \frac{2 \cdot 1}{2} z^2 \\ &= 1 + 2z + z^2. \end{aligned}$$

Imp point:

- In case n is a positive integer, the no. of terms in the expanded form is " $n+1$ ".
- If n is negative or a fraction, then the number of terms in expansion is infinite.

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Binomial Theorem for Positive indices.

$$(a+b)^n = {}^nC_0 a^n b^0 + {}^nC_1 a^{n-1} b^1 + {}^nC_2 a^{n-2} b^2 + {}^nC_3 a^{n-3} b^3 + {}^nC_4 a^{n-4} b^4 + \dots$$

$$\Rightarrow \text{put } n = 2$$
$$(a+b)^2 = {}^2C_0 a^2 b^0 + {}^2C_1 a^{2-1} b^1 + {}^2C_2 a^{2-2} b^2$$
$$= \frac{2!}{0!(2-0)!} a^2 + \frac{2!}{1! \times 1!} a^1 b^1 + \frac{2!}{2! \times 0!} a^0 b^2$$

$$= \frac{2}{1 \times 2} a^2 + \frac{2}{1 \times 1} ab + \frac{2}{2 \times 1} b^2$$
$$= a^2 + 2ab + b^2$$

$$\Rightarrow \text{put } n = 3$$
$$(a+b)^3 = {}^3C_0 a^3 b^0 + {}^3C_1 a^{3-1} b^1 + {}^3C_2 a^{3-2} b^2 + {}^3C_3 a^{3-3} b^3$$
$$= \frac{3!}{0! \times 3!} a^3 + \frac{3!}{1! \times 2!} a^2 b + \frac{6}{1 \times 2} ab^2 + \frac{6}{6 \times 1} b^3$$

$$= a^3 + 3a^2 b + 3ab^2 + b^3$$

$$\Rightarrow \text{put } n = 4$$
$$(a+b)^4 = {}^4C_0 a^4 b^0 + {}^4C_1 a^{4-1} b^1 + {}^4C_2 a^{4-2} b^2 + {}^4C_3 a^{4-3} b^3 + {}^4C_4 a^{4-4} b^4$$
$$= \frac{4!}{0! \times 4!} a^4 + \frac{4!}{1! \times 3!} a^3 b + \frac{4!}{2! \times 2!} a^2 b^2$$

$$+ \frac{4!}{3! \times 1!} ab^3 + \frac{4!}{4! \times 0!} b^4$$
$$= a^4 + 4a^3 b + 6a^2 b^2 + 4ab^3 + b^4$$

Function

$y = x^2$
y depends on x
independent variable
or input
dependent variable
or output

$$y = x^2$$

$$x = 0; y = 0$$

$$x = 2; y = 4$$

$$x = 3; y = 9$$

$$x = 4; y = 16$$

$$\therefore y = f(x)$$

$\Rightarrow y$ is function of x .

A function is a relationship in which for each value of input we get a single & definite value of output.

y is said to be function of x if for each value of x we have a single definite value of y .

Limit of a function

$$\rightarrow y = \frac{x^2 - 4}{x - 2}$$

$$\text{put } x = 2$$

$$y = \frac{2^2 - 4}{2 - 2}$$

$$y = \frac{4-4}{2-2}$$

$$y = \frac{0}{0} \text{ (Indefinite) .}$$

Sol

$$y = \frac{x^2 - 4}{x - 2}$$

$$y = \frac{x^2 - 2^2}{x - 2}$$

$$y = \frac{(x+2)(\cancel{x-2})}{(\cancel{x-2})}$$

$$y = x + 2$$

$$\therefore y = 2 + 2 = 4$$

This method is wrong as we can't cancel 0 by 0

i.e. $(x-2)$ by $(x-2)$ when $x=2$.

$$\rightarrow y = \frac{x^2 - 9}{x - 3}$$

$$\text{If } x = 3$$

$$y = \frac{3^2 - 9}{3 - 3} = \frac{9 - 9}{3 - 3} = \frac{0}{0} \text{ (Indefinite)}$$

$$\Rightarrow y = \frac{x^2 - 3^2}{(x - 3)} = \frac{(x+3)(\cancel{x-3})}{(\cancel{x-3})} = x + 3$$

$$y = 3 + 3 = 6$$

This method is wrong.

limit of a function

$$y = \frac{x^2 - 16}{x - 4}$$

put $x = 4$

$$y = \frac{4^2 - 16}{4 - 4} = \frac{16 - 16}{4 - 4} = \frac{0}{0} \text{ (Indefinite)}$$

Solution

$$y = \frac{x^2 - 16}{x - 4}$$

$$y = \frac{x^2 - 4^2}{x - 4}$$

$$y = \frac{(x + 4)(\cancel{x - 4})}{(\cancel{x - 4})}$$

$$y = x + 4$$

put $x = 4$
 $y = 4 + 4 = 8$

this method is wrong

Actual Method
Limit

$$y = \lim_{x \rightarrow 4} \frac{x^2 - 16}{x - 4}$$

$$y = \lim_{x \rightarrow 4} \frac{x^2 - 4^2}{x - 4}$$

$$y = \lim_{x \rightarrow 4} \frac{(x + 4)(\cancel{x - 4})}{(\cancel{x - 4})}$$

$$y = \lim_{x \rightarrow 4} (x+4)$$

$$y = 4 + 4 = 8$$

$$y = \frac{x^2 - 4}{x - 2}$$

$$\text{At } x = 0$$

$$y = \frac{0}{0} \text{ (Indefinite)}$$

$$y = \lim_{x \rightarrow 2} \frac{x^2 - 4}{x - 2}$$

$$y = \lim_{x \rightarrow 2} \frac{x^2 - 2^2}{x - 2}$$

$$y = \lim_{x \rightarrow 2} \frac{(x+2)(\cancel{x-2})}{\cancel{x-2}}$$

$$y = \lim_{x \rightarrow 2} x + 2$$

$$\text{put } x = 2$$

$$y = 2 + 2 = 4$$

Differentiation.

$$y = x^3$$

Δx = change in x .

Δy = change in y .

$$\frac{\Delta y}{\Delta x} = \frac{\text{change in } x}{\text{change in } y} = \frac{\text{change in output}}{\text{change in input}}$$

If the change in x is very very small then change in y is also very very small.

$$\lim_{x \rightarrow 0} \frac{\Delta y}{\Delta x}$$

$$= \frac{dy}{dx} \rightarrow \text{differentiation / derivative of } y \text{ w.r.t } x.$$

$\frac{dy}{dx} \rightarrow$ Rate of change of y with respect to change in x .

It implies how fast y changes due to change in x .

$$y = x^2$$

$x = 1$	$y = 1$
$x = 2$	$y = 4$
$x = 3$	$y = 9$
$x = 4$	$y = 16$
$x = 5$	$y = 25$

$$y = x^3$$

$x = 1$	$y = 1$
$x = 2$	$y = 8$
$x = 3$	$y = 27$
$x = 4$	$y = 64$
$x = 5$	$y = 125$

Max change:

Examples.

$$\begin{aligned} * \quad \frac{d(x^n)}{dx} &= nx^{n-1} \quad (i) \quad \frac{d}{dx}(x^9) = 9x^8 \\ * \quad \frac{d(cx^n)}{dx} &= c \frac{d}{dx}(x^n) \quad (ii) \quad \frac{d}{dx}(x^{\frac{3}{2}}) = \frac{3}{2}x^{\frac{1}{2}} \\ * \quad \frac{d(c)}{dx} &= 0. \quad (iii) \quad \frac{d}{dx}(10x^3) \\ &= 10 \frac{d}{dx}(x^3) = 10 \cdot 3x^2 \\ &= 30x^2. \end{aligned}$$

$$(iv) \quad \frac{d}{dx}\left(\frac{1}{4}x^4\right) = \frac{1}{4} \cdot 4x^3 = x^3.$$

$$(v) \quad \frac{d}{dx}(100) = 0.$$

~~Examples~~ ~~Examples~~.

$$(vi) \quad \frac{d}{dx}(x^7) = 7x^6.$$

$$\begin{aligned} (vii) \quad \frac{d}{dx}(\sqrt{x}) &= \frac{d}{dx}(x^{\frac{1}{2}}) = \frac{1}{2}x^{-\frac{1}{2}} = \frac{1}{2x^{\frac{1}{2}}} \\ &= \frac{1}{2\sqrt{x}}. \end{aligned}$$

$$(viii) \quad \frac{d}{dx}(100x^3) = 100 \cdot 3x^2 = 300x^2.$$

$$(ix) \quad \frac{d}{dx}\left(\frac{5}{7}x^7\right) = \frac{5}{7} \cdot 7x^6 = 5x^6.$$

$$x \quad \frac{d}{dx}(50) = 0.$$

Q. If $y = \frac{3}{2}x^4$, find $\frac{dy}{dx}$.

$$\frac{dy}{dx} = \frac{d}{dx} \left(\frac{3}{2}x^4 \right)$$

$$\frac{dy}{dx} = \frac{3}{2} \frac{d}{dx} (x^4) = \frac{3}{2} \cdot 4x^3 = 6x^3.$$

Q. If $y = \frac{1}{2}\sqrt{x}$, find $\frac{dy}{dx}$.

$$\frac{dy}{dx} = \frac{d}{dx} \left(\frac{1}{2}\sqrt{x} \right) = \frac{1}{2} \frac{d}{dx} (\sqrt{x})$$

$$= \frac{1}{2} \cdot \frac{1}{2} x^{-\frac{1}{2}}$$

$$= \frac{1}{4} x^{-\frac{1}{2}}$$

$$= \frac{1}{4x^{\frac{1}{2}}} = \frac{1}{4\sqrt{x}}$$

Q. If $y = \frac{3}{x}$, find $\frac{dy}{dx}$.

$$\frac{dy}{dx} = \frac{d}{dx} (3x^{-1})$$

$$= 3 \frac{d}{dx} (x^{-1}) = 3 \times (-x)^{-2}$$

$$= -3x^{-2}$$

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* $\frac{d}{dx} [f_1(x) + f_2(x) + f_3(x) + \dots]$
 $= \frac{d}{dx} (f_1(x)) + \frac{d}{dx} (f_2(x)) + \frac{d}{dx} (f_3(x))$

Q If $y = x^3 + x^2 + 5$, find $\frac{dy}{dx}$

$$\frac{dy}{dx} = \frac{d}{dx} (x^3 + x^2 + 5)$$

$$= \frac{d}{dx} (x^3) + \frac{d}{dx} (x^2) + \frac{d}{dx} (5)$$

$$= 3x^2 + 2x + 0$$

$$\frac{dy}{dx} = 3x^2 + 2x$$

Q If $y = \sqrt{x} + \frac{1}{\sqrt{x}}$, find $\frac{dy}{dx}$

$$\frac{dy}{dx} = \frac{d}{dx} (x^{\frac{1}{2}} + x^{-\frac{1}{2}})$$

$$= \frac{d}{dx} (x^{\frac{1}{2}}) + \frac{d}{dx} (x^{-\frac{1}{2}})$$

$$= \frac{1}{2} x^{-\frac{1}{2}} + \left(-\frac{1}{2}\right) x^{-\frac{3}{2}}$$

$$= \frac{1}{2x^{\frac{1}{2}}} - \frac{1}{2} x^{-\frac{3}{2}}$$

$$= \frac{1}{2\sqrt{x}} - \frac{1}{2x^{\frac{3}{2}}}$$

$$\begin{array}{l} -\frac{1}{2} - \frac{1}{1} \\ \frac{-1-2}{2} = -\frac{3}{2} \\ -\frac{1}{2} - \frac{1}{1} \\ -\left(\frac{1}{2} + \frac{1}{1}\right) \\ \frac{-1+2}{2} = \frac{1}{2} \end{array}$$

$$* \quad \frac{d(x)}{dx} = \frac{d(x)^1}{dx} = 1 \times x^{1-1} = 1x^0 = 1.$$

$$* \quad \frac{d(\sin x)}{dx} = \cos x.$$

$$* \quad \frac{d(\cos x)}{dx} = -\sin x.$$

Q. If $y = x^4 + 2\sin x - 3\cos x$, find $\frac{dy}{dx}$.

$$\frac{dy}{dx} = \frac{d}{dx}(x^4 + 2\sin x - 3\cos x).$$

$$= \frac{d}{dx}(x^4) + \frac{d}{dx}(2\sin x) - \frac{d}{dx}(3\cos x)$$

$$= 4x^3 + 2\frac{d}{dx}(\sin x) - 3\frac{d}{dx}(\cos x)$$

$$= 4x^3 + 2\cos x - 3(-\sin x).$$

$$\frac{dy}{dx} = 4x^3 + 2\cos x + 3\sin x.$$

Q. If $y = \frac{1}{3}x^3 + \frac{1}{2}\sin x + \frac{1}{5}\cos x$, find $\frac{dy}{dx}$.

$$\frac{dy}{dx} = \frac{d}{dx}\left(\frac{1}{3}x^3 + \frac{1}{2}\sin x + \frac{1}{5}\cos x\right).$$

$$= \frac{d}{dx}\left(\frac{1}{3}x^3\right) + \frac{d}{dx}\left(\frac{1}{2}\sin x\right) + \frac{d}{dx}\left(\frac{1}{5}\cos x\right)$$

$$= \frac{1}{3}\frac{d}{dx}(x^3) + \frac{1}{2}\frac{d}{dx}(\sin x) + \frac{1}{5}\frac{d}{dx}(\cos x).$$

$$= \cancel{d} \frac{1}{3} \times 3x^2 + \frac{1}{2} \cos x + \frac{1}{5} (-\sin x)$$

$$= x^2 + \frac{1}{2} \cos x - \frac{1}{5} \sin x$$

Q. If $y = \frac{1}{5}x^5 + 3\cos x - 2$, find $\frac{dy}{dx}$.

$$\frac{dy}{dx} = \frac{d}{dx} \left(\frac{1}{5}x^5 \right) + \frac{d}{dx} (3\cos x) - \frac{d}{dx} (2)$$

$$= \frac{1}{5} \frac{d}{dx} (x^5) + 3 \frac{d}{dx} (\cos x) - 0$$

$$= \frac{1}{5} \times 5x^4 + 3(-\sin x)$$

$$= x^4 - 3\sin x$$

Q. If $y = \frac{3}{x^4} + 10x^3 - \cos x$, find $\frac{dy}{dx}$.

$$\frac{dy}{dx} = \frac{d}{dx} (3x^{-4} + 10x^3 - \cos x)$$

$$= \frac{d}{dx} (3x^{-4}) + \frac{d}{dx} (10x^3) - \frac{d}{dx} (\cos x)$$

$$= 3 \frac{d}{dx} (x^{-4}) + 10 \frac{d}{dx} (x^3) - (-\sin x)$$

$$= 3x(-4x^{-5}) + 10 \times 3x^2 + \sin x$$

$$= -12x^{-5} + 30x^2 + \sin x$$

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Q

$$\text{If } y = x + \sin x + \cos x + 1$$

$$\begin{aligned} \frac{dy}{dx} &= \frac{d}{dx} (x + \sin x + \cos x + 1) \\ &= \frac{d}{dx} (x) + \frac{d}{dx} (\sin x) + \frac{d}{dx} (\cos x) + \frac{d}{dx} (1) \end{aligned}$$

$$= 1 + \cos x - \sin x$$

* $\frac{d}{dx} (\log x) = \frac{1}{x}$

* $\frac{d}{dx} (e^{ax}) = ae^{ax}$
 e.g; $\frac{d}{dx} (e^{3x}) = 3e^{3x}$

* $\frac{d}{dx} (e^x) = e^x$

Q. If $y = 5 \sin x - x^5 + 4 \log x$, find $\frac{dy}{dx}$

$$\frac{dy}{dx} = \frac{d}{dx} (5 \sin x - x^5 + 4 \log x)$$

$$= \frac{d}{dx} (5 \sin x) - \frac{d}{dx} (x^5) + \frac{d}{dx} (4 \log x)$$

$$= 5 \cos x - 5x^4 + \frac{4}{x}$$

$$4x^3 + 10\sin x - 5\cos x + \frac{3}{x} - 2e^x$$

Q

if $y = \frac{1}{2}x^2 + x + \cos x - e^x$.
find $\frac{dy}{dx}$.

$$\begin{aligned}\frac{dy}{dx} &= \frac{d}{dx} \left(\frac{1}{2}x^2 + x + \cos x - e^x \right) \\ &= \frac{d}{dx} \left(\frac{1}{2}x^2 \right) + \frac{d}{dx} (x) + \frac{d}{dx} (\cos x) \\ &\quad - \frac{d}{dx} (e^x).\end{aligned}$$

$$= \frac{1}{2} \times 2x^1 + 1 \times x^0 + (-\sin x) - e^x.$$

$$= x^1 + 1 - \sin x - e^x.$$

Q.

if $y = x^4 + 10\cos x - 5\sin x + 3\log x - 2e^x$
find $\frac{dy}{dx}$.

$$\frac{dy}{dx} = \frac{d}{dx} \left(x^4 + 10\cos x - 5\sin x + 3\log x - 2e^x \right).$$

$$\begin{aligned}&= \frac{d}{dx} (x^4) + \frac{d}{dx} (10\cos x) - \frac{d}{dx} (5\sin x) \\ &\quad + \frac{d}{dx} (3\log x) - \frac{d}{dx} (2e^x)\end{aligned}$$

$$= 4x^3 + 10(-\sin x) - 5\cos x + \frac{3}{x} - 2e^x.$$

$$= 4x^3 - 10\sin x - 5\cos x + \frac{3}{x} - 2e^x$$

Rules for differentiation.

1

Product Rule:

$$\frac{d}{dx}(uv) = u \frac{d}{dx}(v) + v \frac{d}{dx}(u).$$

e.g; If $y = x^3 \sin x$, find $\frac{dy}{dx}$.

$$\frac{dy}{dx} = x^3 \frac{d}{dx}(\sin x) + \sin x \frac{d}{dx}(x^3)$$

$$= x^3 \cdot \cos x + \sin x \cdot 3x^2.$$

$$= x^3 \cos x + 3x^2 \sin x.$$

$$= x^2 (x \cos x + 3 \sin x).$$

e.g; If $y = x^3 \log x$, find $\frac{dy}{dx}$.

$$\frac{dy}{dx} = \frac{d}{dx}(x^3 \log x)$$

$$\frac{dy}{dx} = x^3 \frac{d}{dx}(\log x) + \log x \frac{d}{dx}(x^3)$$

$$\frac{dy}{dx} = x^3 \times \frac{1}{x} + \log x \times 3x^2.$$

$$\frac{dy}{dx} = x^2 + \log x \cdot 3x^2$$

$$\frac{dy}{dx} = x^2 + 3x^2 \log x = x^2 (1 + 3 \log x)$$

→

If $y = x \sin x$, find $\frac{dy}{dx}$.

$$\begin{aligned}\frac{dy}{dx} &= \frac{d}{dx}(x \sin x) \\ &= x \frac{d}{dx}(\sin x) + \sin x \left(\frac{d}{dx}(x) \right) \\ &= x \times (\cos x) + \sin x \times 1 \\ &= x \cos x + \sin x.\end{aligned}$$

→

If $y = x \log x$

$$\begin{aligned}\frac{dy}{dx} &= \frac{d}{dx}(x \log x) \\ &= x \frac{d}{dx}(\log x) + \log x \cdot \frac{d}{dx}(x) \\ &= x \times \frac{1}{x} + \log x \times 1 \\ &= 1 + \log x.\end{aligned}$$

Quotient Rule.

$$\frac{d}{dx} \left(\frac{u}{v} \right) = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}.$$

e.g. If $y = \frac{\sin x}{x}$, find $\frac{dy}{dx}$.

$$\begin{aligned}\frac{d}{dx} \left(\frac{\sin x}{x} \right) &= \frac{x \frac{d}{dx}(\sin x) - \sin x \frac{d}{dx}(x)}{x^2} \\ &= \frac{x \cos x - \sin x \times 1}{x^2}\end{aligned}$$

$\frac{p}{B}$

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Derivative / Differentiation of $\tan x$.

$$y = \tan x$$

$$\frac{dy}{dx} = \frac{d(\tan x)}{dx}$$

$$= \frac{d}{dx} \left(\frac{\sin x}{\cos x} \right)$$

$$= \frac{\cos x \left(\frac{d}{dx} (\sin x) \right) - \sin x \frac{d}{dx} (\cos x)}{\cos^2 x}$$

$$= \frac{\cos x \times \cos x - \sin x (-\sin x)}{\cos^2 x}$$

$$= \frac{\cos^2 x + \sin^2 x}{\cos^2 x} = \frac{1}{\cos^2 x}$$

$$= \frac{1}{\cos x} \times \frac{1}{\cos x} = \sec x \times \sec x$$

$$\therefore \frac{dy}{dx} = \sec^2 x$$

Derivative of $\cot x$.

$$y = \cot x.$$

$$\frac{dy}{dx} = \frac{d}{dx} (\cot x).$$

$$= \frac{d}{dx} \left(\frac{\cos x}{\sin x} \right)$$

$$= \frac{\sin x \left(\frac{d}{dx} (\cos x) \right) - \cos x \frac{d}{dx} (\sin x)}{\sin^2 x}$$

$$= \frac{\sin x (-\sin x) - \cos x \times \cos x}{\sin^2 x}$$

$$= \frac{-\sin^2 x - \cos^2 x}{\sin^2 x}$$

$$= \frac{-(\sin^2 x + \cos^2 x)}{\sin^2 x} = \frac{-1}{\sin^2 x}$$

$$= -\operatorname{cosec}^2 x.$$

$$\therefore \frac{d}{dx} (\cot x) = -\operatorname{cosec}^2 x.$$

Differentiation of $\sec x$.

$$y = \sec x.$$

$$\frac{dy}{dx} = \frac{d(\sec x)}{dx} \\ = \frac{d\left(\frac{1}{\cos x}\right)}{dx}$$

$$\frac{dy}{dx} = \frac{\cos x \cdot \frac{d}{dx}(1) - 1 \cdot \frac{d}{dx}(\cos x)}{\cos^2 x}$$

$$= \frac{\cos x \cdot 0 - 1 \times (-\sin x)}{\cos^2 x}$$

$$= \frac{0 + \sin x}{\cos^2 x} = \frac{\sin x}{\cos^2 x}$$

$$\frac{dy}{dx} = \frac{1}{\cos} \cdot \frac{\sin x}{\cos x}$$

$$= \sec x \tan x.$$

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Differentiation of cosec x.

$$y = \text{cosec } x.$$

$$\frac{dy}{dx} = \frac{d}{dx} (\text{cosec } x).$$

$$= \frac{d}{dx} \left(\frac{1}{\sin x} \right).$$

$$= \frac{\sin x \frac{d}{dx} (1) - 1 \frac{d}{dx} (\sin x)}{\sin^2 x}$$

$$= \frac{\sin x \times 0 - 1 \times (\cos x)}{\sin^2 x}$$

$$= \frac{0 - \cos x}{\sin^2 x} = - \frac{\cos x}{\sin^2 x}$$

$$= - \frac{1}{\sin x} \cdot \frac{\cos x}{\sin x}$$

$$= - \text{cosec } x \cdot \cot x.$$

3

Q Parametric form.

$y = x^3 + x^2$ and $x = x^4 + x^3$, where x is the parameter. find $\frac{dy}{dx}$.

$$y = x^3 + x^2$$

$$\therefore \frac{dy}{dx} = \frac{d}{dx}(x^3 + x^2)$$

$$= \frac{d}{dx}(x^3) + \frac{d}{dx}(x^2)$$

$$= 3x^2 + 2x$$

$$x = x^4 + x^3$$

$$\therefore \frac{dx}{dx} = \frac{d}{dx}(x^4 + x^3)$$

$$= \frac{d}{dx}(x^4) + \frac{d}{dx}(x^3)$$

$$= 4x^3 + 3x^2$$

$$\therefore \frac{dy}{dx} = \frac{dy/dx}{dx/dx}$$

$$= \frac{(3x^2 + 2x)}{(4x^3 + 3x^2)}$$

$$= \frac{\cancel{x}(3\cancel{x} + 2)}{\cancel{x}(4\cancel{x}^2 + 3\cancel{x})} = \frac{3x + 2}{4x^2 + 3x}$$

Q If $y = a \cos \theta$ and $x = b \sin \theta$
where θ is the parameter, find $\frac{dy}{dx}$

$$y = a \cos \theta$$

$$\frac{dy}{d\theta} = \frac{d(a \cos \theta)}{d\theta}$$

$$= a \frac{d(\cos \theta)}{d\theta}$$

$$= a \times (-\sin \theta)$$

$$= -a \sin \theta$$

$$x = b \sin \theta$$

$$\frac{dx}{d\theta} = \frac{d(b \sin \theta)}{d\theta}$$

$$= b \frac{d(\sin \theta)}{d\theta}$$

$$= b \times \cos \theta = b \cos \theta$$

$$\frac{dy}{dx} = \frac{dy/d\theta}{dx/d\theta}$$

$$= \frac{-a \sin \theta}{b \cos \theta} = -\frac{a}{b} \tan \theta$$

4 Logarithmic form:

$$\rightarrow \frac{d}{dx} (\log x) = \frac{1}{x} \times 1 = \frac{1}{x}$$

$$\rightarrow \frac{d}{dx} (\log 5x) = \frac{1}{5x} \times 5 = \frac{1}{x}$$

$$\rightarrow \frac{d}{dx} [\log(\sin x)] = \frac{1}{\sin x} \times \cos x$$

$$\rightarrow \frac{d}{dx} [\sin(\log x)] = \cos(\log x) \times \frac{1}{x}$$
$$= \frac{\cos(\log x)}{x}$$

$$\rightarrow \frac{d}{dx} [\log(x^3)] = \frac{1}{x^3} \times 3x^2$$
$$= \frac{3}{x}$$

*

$$\frac{d}{dx} (\log y) = \frac{1}{y} \cdot \frac{dy}{dx}$$

$$\log m^n = n \log m$$
$$\Rightarrow \log x^3 = 3 \log x$$

$$y = a^x.$$

Taking log on b/s

$$\log y = \log a^x.$$

$$\log y = x \log a.$$

to
differentiating b/s w.r.t. x ,

$$\frac{d}{dx} (\log y) = \frac{d}{dx} (x \log a).$$

$$\frac{1}{y} \frac{dy}{dx} = \log a \frac{d}{dx} (x).$$

$$\frac{1}{y} \frac{dy}{dx} = \log a \cdot 1.$$

$$\frac{dy}{dx} = y \log a.$$

$$\frac{dy}{dx} = a^x \log a.$$

$$\frac{d(a^x)}{dx} = a^x \log a.$$

$$\text{eg } \frac{d}{dx} (100^x) = 100^x \log 100.$$

$$y = x^x$$

$$\log y = \log x^x$$

$$\log y = x \log x$$

Differentiating b/s w.r.t. x

$$\frac{d(\log y)}{dx} = \frac{d}{dx} (x \log x)$$

$$\frac{1}{y} \cdot \frac{dy}{dx} = x \frac{d}{dx} (\log x) + \log x \frac{d}{dx} (x)$$

$$\frac{1}{y} \cdot \frac{dy}{dx} = x \cdot \frac{1}{x} + \log x \cdot 1$$

$$\frac{1}{y} \cdot \frac{dy}{dx} = (1 + \log x)$$

$$\frac{dy}{dx} = y(1 + \log x)$$

$$\frac{dy}{dx} = x^x (1 + \log x)$$

$$y = x^{\sin x}$$

$$\log y = \log x^{\sin x}$$

$$\log y = \sin x \log x$$

$$\log y = \sin x \log x$$

$\sin(3x^2+2x)$

$$\frac{d}{dx}(\log y) = \frac{d}{dx}(\sin x \log x).$$

$$\frac{1}{y} \frac{dy}{dx} = \sin x \frac{d}{dx}(\log x) + \log x \frac{d}{dx}(\sin x)$$

$$\frac{1}{y} \frac{dy}{dx} = \sin x \times \frac{1}{x} + \log x \times \cos x.$$

$$\frac{1}{y} \frac{dy}{dx} = \frac{\sin x}{x} + \log x \times \cos x.$$

$$\frac{dy}{dx} = y \left(\frac{\sin x}{x} + \cos x \times \log x \right).$$

$$= x^{\sin x} \left(\frac{\sin x}{x} + \cos x \cdot \log x \right).$$

5

Chain Rule:

composite function: It is a function which is composed of many functions.

e.g; $\rightarrow y = (x^3 + x^2)^4$

$\rightarrow y = (\sin x + \cos x)^3$

$\rightarrow y = \log(\sin x)$

$\rightarrow y = \sin(3x^2 + 2)^3$

*

Chain rule is used to find the derivative of a composite function.

i)

$$y = (x^4 + x^3)^5$$

$$\frac{dy}{dx} = 5(x^4 + x^3)^4 \times 4x^3 + 3x^2$$

$$(ii) \quad y = (x^6 + 4)^{10}$$

$$\begin{aligned} \frac{dy}{dx} &= 10(x^6 + 4)^9 \times (6x^5 + 0) \\ &= 10(x^6 + 4)^9 \cdot 6x^5 \\ &= 60x^5 (x^6 + 4)^9 \end{aligned}$$

$$(iii) \quad y = (\sin x + \cos x)^3$$

$$\frac{dy}{dx} = 3(\sin x + \cos x)^2 \times (\cos x - \sin x)$$

$$(iv) \quad y = \log(x^3 + x^2)$$

$$\frac{dy}{dx} = \frac{1}{x^3 + x^2} \times (3x^2 + 2x)$$

$$= \frac{3x^2 + 2x}{x^3 + x^2} = \frac{x(3x + 2)}{x^2(x + 1)}$$

$$= \frac{3x + 2}{x^2 + x}$$

$$\frac{3x^2}{3x^2}$$

$$(v) \quad y = \sin(x^3 + 3x^2)$$

$$\frac{dy}{dx} = \cos(x^3 + 3x^2) \times (3x^2 + 6x)$$

$$(vi) \quad y = \sin(10x)$$

$$\frac{dy}{dx} = \cos 10x \times 10$$

$$= 10 \cos 10x$$

(vii) y

no int
(viii).

no int
IX



Q

(vii)

$$y = \log 5x.$$

$$\frac{dy}{dx} = \frac{1}{5x} \times 5 = \frac{1}{x}.$$

(viii)

$$y = \sin ax.$$

$$\frac{dy}{dx} = \cos ax \times a = a \cos ax.$$

e.g; $y = \sin 2x.$

$$\frac{dy}{dx} = 2 \cos 2x.$$

$$y = \cos ax.$$

$$\frac{dy}{dx} = -\sin ax \times a = -a \sin ax.$$

e.g; $y = \cos 2x = -2 \sin 2x.$

calculation of instantaneous speed.
or instantaneous velocity.

(The speed or velocity of a particle at any instant of time).

Q

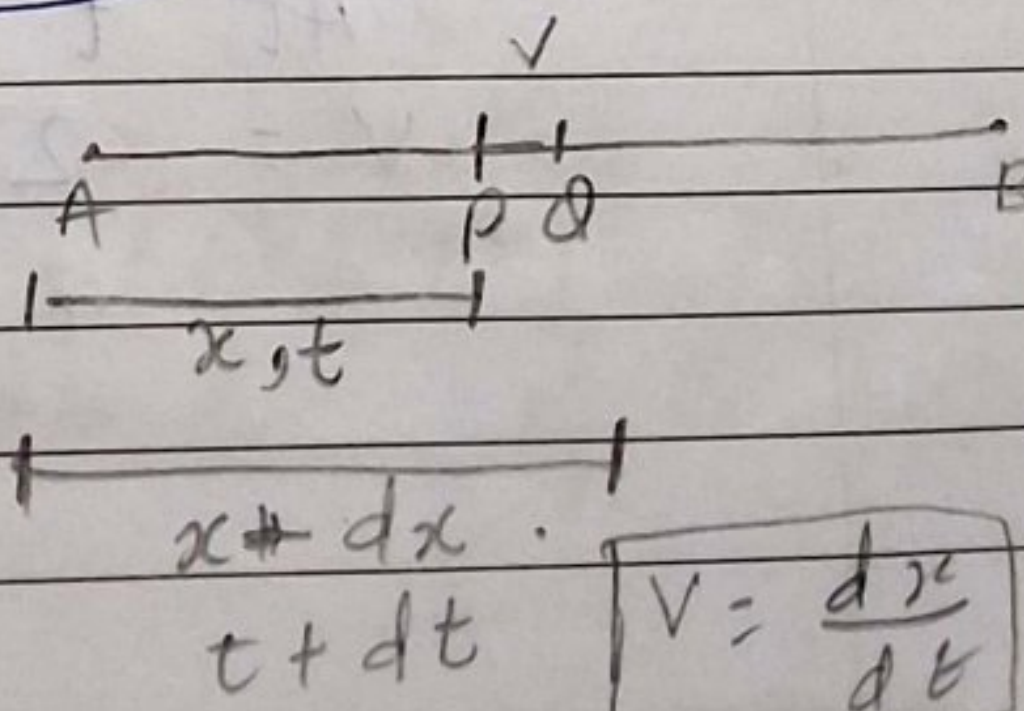
The distance travelled by a particle is given by

$$x = (t^3 + 5) \text{ m}.$$

find speed of particle at $t = 2 \text{ sec}.$

$$x = t^3 + 5.$$

$$v = \frac{dx}{dt}$$



$$v = \frac{d}{dx}(t^3 + 5) = \frac{d}{dx}(t^3) + \frac{d}{dx}(5)$$

$$= 3t^2 + 0$$

$$= 3t^2$$

Now put $t = 2$

we get

$$v = 3 \times (2)^2 = 3 \times 4 = 12 \text{ m/s}$$

Q

The displacement of a particle is given by $x = (t^2 + 5t + 2) \text{ m}$. Find the initial velocity of the particle and its velocity at $t = 3 \text{ sec}$.

$$v = \frac{dx}{dt}$$

$$= \frac{d}{dt}(t^2 + 5t + 2)$$

$$v = \frac{d}{dt}(t^2) + \frac{d}{dt}(5t) + \frac{d}{dt}(2)$$

$$= 2t + 5 \times 1 + 0$$

$$= 2t + 5$$

Initial velocity

$t = 0 \text{ sec}$

$$v = 2(0) + 5 = 5 \text{ m/s}$$

At $t = 3 \text{ sec}$

$$v = 2(3) + 5 = 6 + 5 = 11 \text{ m/s}$$

Q The distance travelled by a particle is given by $x = (2t + 10)$ m. Show that the particle is moving with constant speed.

$$x = (2t + 10)$$

$$v = \frac{dx}{dt} = \frac{d(2t + 10)}{dt}$$

$$v = \frac{d(2t)}{dt} + \frac{d(10)}{dt}$$

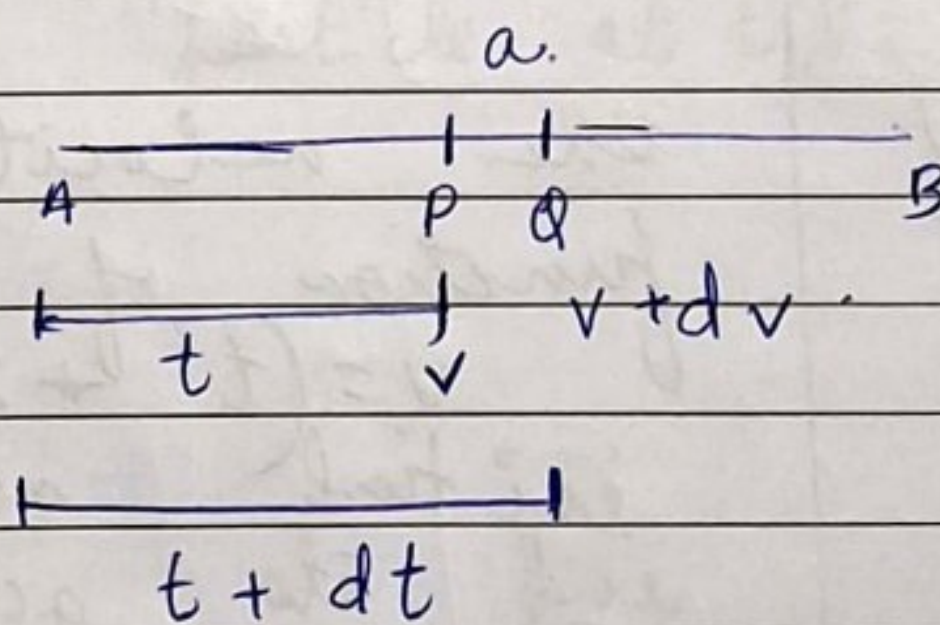
$$= 2 \cdot 1 + 0 = 2 \text{ m/sec}$$

clearly speed of the particle is constant as v is independent of time.

calculation of Instantaneous acceleration (acceleration of a particle at any instant).

$$a = \frac{v + dv - v}{t + dt - t}$$

$$a = \frac{dv}{dt}$$



Q The velocity of a particle is given by $v = (t^3 + t^2)$ m/s. Find the acceleration of the particle at $t = 3$ s.

$$a = \frac{dv}{dt}$$

$$= \frac{d(t^3 + t^2)}{dt}$$

$$= \frac{d(t^3)}{dt} + \frac{d(t^2)}{dt}$$

$$a = (3t^2 + 2t) \text{ m/s}^2$$

Now at $t = 3 \text{ sec.}$

$$a = 3 \times 3^2 + 2 \times 3$$

$$= 3 \times 9 + 6$$

$$= 27 + 6$$

$$= 33 \text{ m/s}^2$$

Q

The velocity of a particle as a function of time is given by $v = (t^2 + 2t + 5) \text{ m/s}$. Find the initial acceleration of the particle and its acceleration at $t = 2 \text{ sec.}$

$$a = \frac{dv}{dt}$$

$$a = \frac{d(t^2 + 2t + 5)}{dt}$$

$$a = \frac{d(t^2)}{dt} + \frac{d(2t)}{dt} + \frac{d(5)}{dt}$$

$$a = 2t + 2 + 0$$

$$a = (2t + 2) \text{ m/s}^2$$

Initial acceleration

$$\text{at } t = 0.$$

$$a = 2 \times 0 + 2 = 0 + 2 = 2 \text{ m/s}^2$$

$$\text{at } t = 2 \text{ sec.}$$

$$a = 2 \times 2 + 2 = 6 \text{ m/s}^2$$

Q. The velocity of a particle of mass 5 Kg is given by $v = (t^2 + 10) \text{ m/s}$. Find the force acting on the particle at $t = 2 \text{ sec}$.

$$a = \frac{dv}{dt}$$

$$a = \frac{d(t^2 + 10)}{dt}$$

$$a = \frac{d(t^2)}{dt} + \frac{d(10)}{dt}$$

$$= 2t + 0$$

$$a = 2t \text{ m/s}^2$$

∴ Force at $t = 2 \text{ sec}$.

We know, $F = m \cdot a$.

$$F = 5 \times 2 \times 2 \\ = 5 \times 4 = \underline{20 \text{ N}}.$$

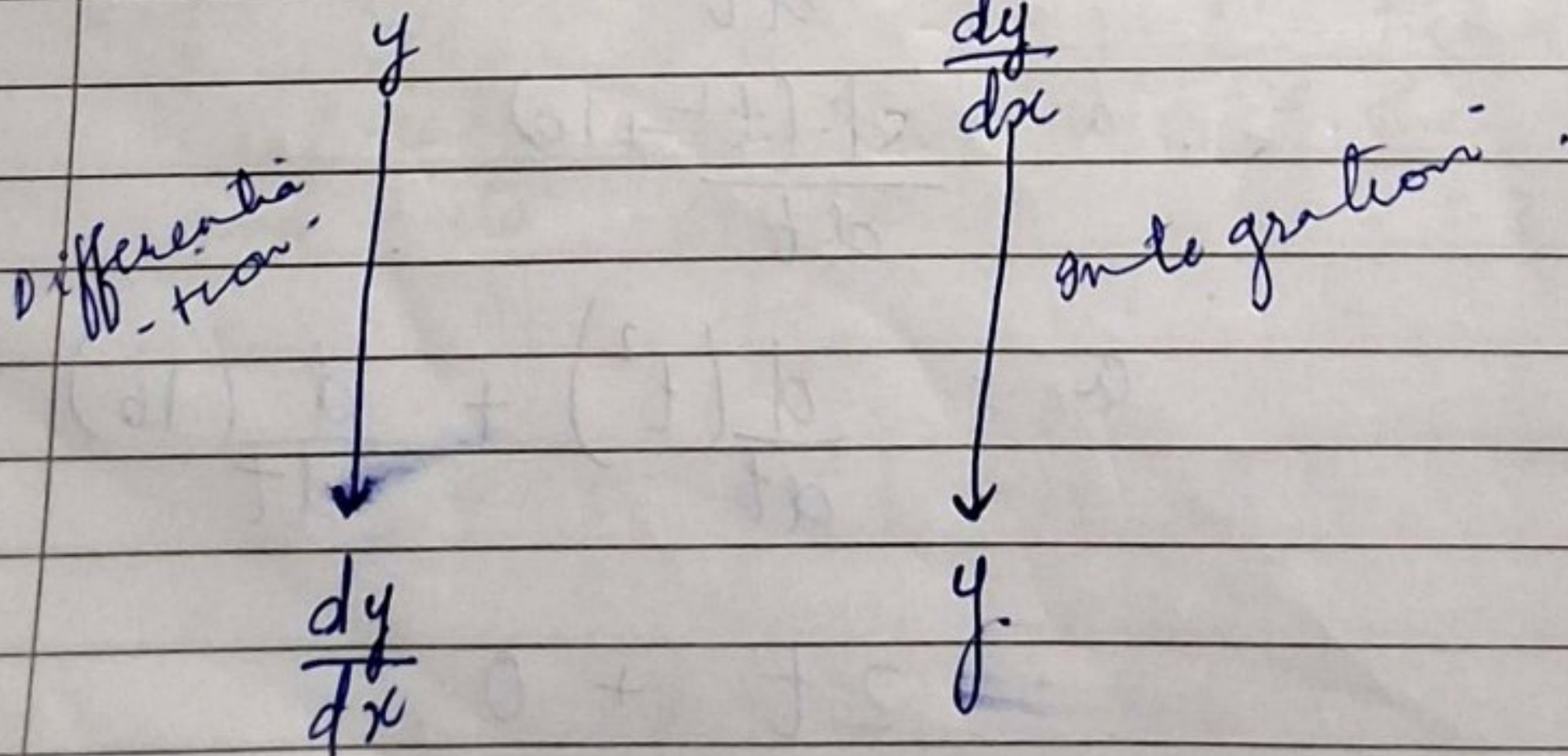
Integration $\int f(x) dx$

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C. \quad (\text{Here } n \neq -1)$$

$$\text{e.g.; } \int x^3 dx = \frac{x^4}{4} + C.$$

$$\int x^{10} dx = \frac{x^{11}}{11} + C.$$

$$\int \frac{1}{x^2} dx = \int x^{-2} dx = \frac{x^{-1}}{-1} + C \\ = -x^{-1} + C = -\frac{1}{x} + C.$$



_ / _ / _

★ Differentiation & integration are
inverse of each other -

y differentiation integration y

Q

Q1 $\frac{dy}{dx} = 4x^3$ find y

$$dy = 4x^3 dx$$

Integrating b/s

$$\int dy = \int 4x^3 dx$$

$$y = 4 \int x^3 dx$$

$$y = \frac{4x^4}{4} + C$$

$$y = x^4 + C$$

Q

Q1 $\frac{dy}{dx} = \frac{5}{2} x^3$ find y

$$dy = \frac{5}{2} x^3 dx$$

Integrating b/s

$$\int dy = \int \frac{5}{2} x^3 dx$$

$$y = \frac{5}{2} \int x^3 dx$$

$$y = \frac{5}{2} \cdot \frac{x^4}{4} + C$$

$$y = \frac{5}{8} x^4 + C$$

Q. If $\frac{dy}{dx} = 3\sqrt{x}$ find y .

$$dy = 3\sqrt{x} dx$$

integrating b/s.

$$\int dy = \int 3\sqrt{x} dx$$

$$y = 3 \int \sqrt{x} dx$$

$$y = 3 \int x^{\frac{1}{2}} dx$$

$$y = \frac{3 \cdot x^{\frac{3}{2}}}{\frac{3}{2}} + C$$

$$y = 2x^{\frac{3}{2}} + C$$

Q If $\frac{dy}{dx} = \frac{7}{3}x^6$ find y .

$$dy = \frac{7}{3}x^6 dx$$

integrating on b/s.

$$\int dy = \int \frac{7}{3}x^6 dx$$

$$y = \frac{7}{3} \int x^6 dx$$

$$y = \frac{7}{3} \cdot \frac{x^7}{7} + C$$

$$y = \frac{x^7}{3} + C$$

Differentiation of constant is always zero but integration of constant is non zero.

$$\int k dx = k \int dx = kx + C$$

e.g. $\int 10 dx = 10 \int dx = 10x + C$

Q If $\frac{dy}{dx} = \frac{3}{2}x^2 + 5x + 3$ find y .

$$dy = \left(\frac{3}{2}x^2 + 5x + 3 \right) dx$$

Integrating b/s.

$$\int dy = \int \left(\frac{3}{2}x^2 + 5x + 3 \right) dx$$

$$y = \int \frac{3}{2}x^2 dx + \int 5x dx + \int 3 dx$$

$$y = \frac{3}{2} \int x^2 dx + 5 \int x dx + 3 \int dx$$

$$y = \frac{3}{2} \frac{x^3}{3} + C_1 + 5 \frac{x^2}{2} + C_2 + 3x + C_3$$

$$y = \frac{1}{2} x^3 + \frac{5}{2} x^2 + 3x + C$$

Q If $\frac{dy}{dx} = \frac{1}{x^3} + 4x + 100$.

$$dy = \left(\frac{1}{x^3} + 4x + 100 \right) dx$$

$$\int dy = \int \left(\frac{1}{x^3} + 4x + 100 \right) dx$$

$$y = \int x^{-3} dx + \int 4x dx + \int 100 dx$$

$$y = \frac{x^{-2}}{-2} + C_1 + \frac{4 \cdot x^2}{2} + C_2 + 100x + C_3$$

$$y = -\frac{x^{-2}}{2} + 2x^2 + 100x + C$$

Q

If $\frac{dy}{dx} = 3x^2 + 2x + 1$

$$dy = (3x^2 + 2x + 1) dx$$

Integrating on both sides.

$$\int dy = \int (3x^2 + 2x + 1) dx$$

$$y = \int 3x^2 dx + \int 2x dx + \int 1 dx$$

$$y = \frac{3 \cdot x^3}{3} + \frac{2x^2}{2} + 1x + C_3$$

$$= x^3 + x^2 + x + C$$

$$* \int \sin x \, dx = -\cos x + C$$

$$* \int \cos x \, dx = \sin x + C$$

Q If $\frac{dy}{dx} = \frac{1}{2}x - 3\cos x$, find y .

$$dy = \left(\frac{1}{2}x - 3\cos x\right) dx$$

$$\int dy = \int \left(\frac{1}{2}x - 3\cos x\right) dx$$

$$y = \int \frac{1}{2}x \, dx - \int 3\cos x \, dx$$

$$y = \frac{1}{2} \cdot \frac{x^2}{2} + C_1 - 3(\sin x + C_2)$$

$$y = \frac{x^2}{4} - 3\sin x + C$$

$$* \int e^{ax} \, dx = \frac{e^{ax}}{a} + C$$

Q If $\frac{dy}{dx} = 5x^3 - 4\sin x + 3$, find y .

$$dy = (5x^3 - 4\sin x + 3) \, dx$$

$$\int dy = \int (5x^3 - 4\sin x + 3) \, dx$$

$$y = \int 5x^3 dx - \int 4 \sin x dx + \int 3 dx$$

$$y = \frac{5x^4}{4} - 4(-\cos x + C_2) + 3x + C_3$$

$$y = \frac{5x^4}{4} + 4 \cos x + 3x + C$$

Q. 2f $\frac{dy}{dx} = \frac{1}{2}x^3 + 4 \cos x - e^{2x}$ find y.

$$dy = \left(\frac{1}{2}x^3 + 4 \cos x - e^{2x} \right) dx$$

$$\Rightarrow \int dy = \int \left(\frac{1}{2}x^3 + 4 \cos x - e^{2x} \right) dx$$

$$y = \int \frac{1}{2}x^3 dx + \int 4 \cos x dx - \int e^{2x} dx$$

$$y = \frac{1}{2} \cdot \frac{x^4}{4} + C_1 + 4 \sin x + C_2 - \frac{e^{2x}}{2} + C_3$$

$$y = \frac{1}{8}x^4 + 4 \sin x - \frac{e^{2x}}{2} + C$$

* $\int x^{-1} dx = \int \frac{1}{x} dx = \log x + C$

Q. If $\frac{dy}{dx} = -\frac{5}{x} + 3 \sin x - 10e^{2x}$, find y

$$dy = \left(-\frac{5}{x} + 3 \sin x - 10e^{2x} \right) dx$$

$$\int dy = \int \left(-\frac{5}{x} + 3 \sin x - 10e^{2x} \right) dx$$

$$y = \int -\frac{5}{x} dx + \int 3 \sin x dx - \int 10e^{2x} dx$$

$$y = 5 \log x + C_1 + 3(-\cos x + C_2) - 10 \frac{e^{2x}}{2} + C_3$$

$$y = 5 \log x - 3 \cos x - 5e^{2x} + C$$

* $\int \sin ax dx = -\frac{\cos ax}{a} + C$

e.g; $\int \sin 2x dx = -\frac{\cos 2x}{2} + C$

* $\int \sin 2\omega t dt = -\frac{\cos 2\omega t}{2\omega} + C$

* $\int \cos ax dx = \frac{\sin ax}{a} + C$

e.g $\int \cos 2x dx = \frac{\sin 2x}{2} + C$

* $\int \cos 2\omega t dt = \frac{\sin 2\omega t}{2\omega} + C$

$$\int f(x) dx$$

Indefinite integration e.g. $\int x^2 dx$.

upper limit

$$\int_a^b f(x) dx$$

lower limit

Definite integration.

e.g. $\int_1^3 x^2 dx$.

⇒ In indefinite integration ($\int f(x) dx$) we don't know the initial and final values of the variable.

⇒ In definite integration ($\int_a^b f(x) dx$) we know the initial and final values of the variable. The initial value is known as lower limit and the final value is known as upper limit.

e.g. $\int_1^3 x^2 dx = \left(\frac{x^3}{3} \right)_1^3$

$$= \frac{1}{3} (x^3)_1^3$$

$$= \frac{1}{3} (3^3 - 1^3) = \frac{1}{3} \times (27 - 1)$$

$$= \frac{1}{3} \times 26 = \frac{26}{3}$$

$$\Rightarrow \int_1^2 x \, dx = \left(\frac{x^2}{2} \right)_1^2 = \frac{1}{2} (x^2)_1^2$$

$$= \frac{1}{2} (2^2 - 1^2)$$

$$= \frac{1}{2} (4 - 1) = \frac{3}{2}$$

$$\Rightarrow \int_2^1 x^2 \, dx = \left(\frac{x^3}{3} \right)_2^1 = \frac{1}{3} (x^3)_2^1$$

$$= \frac{1}{3} (1^3 - 2^3)$$

$$= \frac{1}{3} (1 - 8)$$

$$= \frac{1}{3} \times (-7) = -\frac{7}{3}$$

$$\int_0^3 x^2 \, dx = \left(\frac{x^3}{3} \right)_0^3 = \frac{1}{3} (3^3 - 0^3)$$

$$= \frac{1}{3} \times 27 - 0$$

$$= \frac{27}{3} = 9$$

$$\int_{-1}^1 x \, dx = \left(\frac{x^2}{2} \right)_{-1}^1 = \frac{1}{2} (x^2)_{-1}^{+1}$$

$$= \frac{1}{2} (1^2 - (-1)^2)$$

$$= \frac{1}{2} (1 - 1) = 0$$

$$\begin{aligned}
 \Rightarrow \int_{-1}^2 x^2 dx &= \left(\frac{x^3}{3} \right)_{-1}^2 \\
 &= \frac{1}{3} \left(x^3 \right)_{-1}^2 = \frac{1}{3} (2^3 - (-1)^3) \\
 &= \frac{1}{3} (8 - (-1)) \\
 &= \frac{1}{3} (8 + 1) \\
 &= \frac{9}{3} = 3.
 \end{aligned}$$

$$\begin{aligned}
 \Rightarrow \int_2^{-1} x^3 dx &= \left(\frac{x^4}{4} \right)_{-1}^2 \\
 &= \frac{1}{4} \left(x^4 \right)_{-1}^2 = \frac{1}{4} (2^4 - (-1)^4) \\
 &= \frac{1}{4} (16 - 1) \\
 &= \frac{15}{4}
 \end{aligned}$$

$$\begin{aligned}
 \Rightarrow \int_{-1}^1 x^3 dx &= \left(\frac{x^4}{4} \right)_{-1}^1 \\
 &= \frac{1}{4} \left(x^4 \right)_{-1}^1 = \frac{1}{4} (1^4 - (-1)^4) \\
 &= \frac{1}{4} (1 - 1) = 0.
 \end{aligned}$$

$$\Rightarrow \int_0^{-2} 3x \, dx = \left(\frac{3x^2}{2} \right)_0^{-2}$$

$$\int_0^{-2} 3x \, dx = \frac{3}{2} \left(x^2 \right)_0^{-2} = \frac{3}{2} \left((-2)^2 - (0)^2 \right)$$

$$= \frac{3}{2} (4 - 0)$$

$$= \frac{3}{2} \times 4 = 6$$

$\Rightarrow$ degree

radian

	360°				2π
	180°				π
	90°				$\frac{\pi}{2}$
	60°				$\frac{\pi}{3}$
$0, \frac{1}{2}, \frac{1}{\sqrt{2}}, \frac{\sqrt{3}}{2}, 1$	45°				$\frac{\pi}{4}$
	30°				$\frac{\pi}{6}$
	0°				0
	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$
sin	0	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$	1
cos	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0

* $\sin \pi = 0$, $\sin 2\pi = 0$

$\cos \pi = -1$, $\cos 2\pi = 1$

$\Rightarrow$ Evaluate $\int_0^{\pi/2} \cos x \, dx$

$$\int_0^{\pi/2} \cos x \, dx = \left(\sin x \right)_0^{\pi/2}$$

$$= \sin \frac{\pi}{2} - \sin 0 = 1 - 0 = 1$$

$$\sin n\pi = 0$$

$$n = 0, 1, 2, \dots$$

$$\cos n\pi = -1$$

$$n \text{ is odd}$$

$$\cos n\pi = 1$$

$$n \text{ is even}$$

Evaluate integration of $\sin x$. $\int_0^{\pi/2} \sin x \, dx$

$$\int_0^{\pi/2} \sin x \, dx = (-\cos x)_0^{\pi/2}$$

$$= -(\cos x)_0^{\pi/2}$$

$$= -(\cos \frac{\pi}{2} - \cos 0)$$

$$= -(0 - 1)$$

$$= -(-1) = 1$$

Evaluate $\int_0^{2\pi} \cos x \, dx$.

$$\int_0^{2\pi} \cos x \, dx = (\sin x)_0^{2\pi}$$

$$= \sin 2\pi - \sin 0$$

$$= 0 - 0 = 0$$

~~Evaluate~~ $\int_{\pi/2}^{\pi} \cos \theta \, d\theta = (\sin \theta)_{\pi/2}^{\pi}$

$$= \sin \pi - \sin \frac{\pi}{2}$$

$$= 0 - 1 = -1$$

$$\int_{\pi/2}^{2\pi} \cos \theta \, d\theta = (\sin \theta)_{\pi/2}^{2\pi}$$

$$= \sin 2\pi - \sin \frac{\pi}{2}$$

$$= 0 - 1 = -1$$

//_

V_{avg}^{∞} Evaluate $\int_0^T \cos \omega t dt$, where $\omega = \frac{2\pi}{T}$.

$$\begin{aligned} &= \left(\frac{\sin \omega t}{\omega} \right)_0^T = \frac{1}{\omega} \left(\sin \omega t \right)_0^T \\ &= \frac{1}{\omega} \left(\sin \frac{2\pi}{T} \cdot t \right)_0^T \\ &= \frac{2\pi}{T} \left(\sin \frac{2\pi}{T} \cdot T - \sin 0 \right) \\ &= \frac{T}{2\pi} \left(\sin 2\pi - \sin 0 \right) \\ &= \frac{T}{2\pi} (0 - 0) = 0 \end{aligned}$$

Evaluate $\int_0^{T/2} \cos 2\omega t dt$.

$$\begin{aligned} \int_0^{T/2} \cos 2\omega t dt &= \left(\frac{\sin 2\omega t}{2\omega} \right)_0^{T/2} \\ &= \frac{1}{2\omega} \left(\sin \frac{2\pi}{T} t \right)_0^{T/2} \\ &= \frac{T}{2 \cdot 2\pi} \left(\sin \frac{2\pi}{T} t \right)_0^{T/2} \\ &= \frac{T}{4\pi} \left(\sin \frac{4\pi}{T} t \right)_0^{T/2} = \frac{T}{4\pi} \left(\sin \frac{4\pi \cdot T/2}{T} - \sin 0 \right) \\ &= \frac{T}{4\pi} \left(\sin 2\pi - \sin 0 \right) = \frac{T}{4\pi} (0 - 0) = 0 \end{aligned}$$

Evaluate $\int_u^v du = \int_0^t a dt$

$$\int_u^v du = a \int_0^t dt$$

$$(u)_u^v = a(t)_0^t$$

$$v - u = a(t - 0)$$

$$v - u = at$$

$$v = u + at$$

Evaluate $\int_0^s dx = \int_0^t u dt + \int_0^t at dt$

$$\int_0^s dx = u \int_0^t dt + a \int_0^t t dt$$

$$(x)_0^s = u(t)_0^t + a \left(\frac{t^2}{2} \right)_0^t$$

$$s - 0 = u(t - 0) + \frac{1}{2} a (t^2)_0^t$$

$$s = ut + \frac{1}{2} a (t^2 - 0^2)$$

$$s = ut + \frac{1}{2} at^2$$

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Evaluate $\int_u^v u \, du = \int_0^s a \, dx$

$$\int_u^v u \, du = a \int_0^s dx$$

$$\left(\frac{u^2}{2} \right)_u^v = a (x)_0^s$$

$$\frac{1}{2} (u^2)_u^v = a (s - 0)$$

$$\frac{1}{2} (v^2 - u^2) = as$$

$$v^2 - u^2 = 2as$$